

MEMBUAT JARINGAN OPEN SHORTEST PATH FIRST (OSPF) PADA ROUTER MIKROTIK

Disusun Oleh

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Little Things About Me

- Using MikroTik RouterOS (v5.20) Since 14 December 2014
 - RouterOS x86 at PC
- 24 April 2016, MTCNA #1604NA934
- 31 July 2016, MTCRE #1607RE248
- 3 August 2016, MikroTik Certified Consultant
- Still in School, SMK Taruna Bhakti Depok

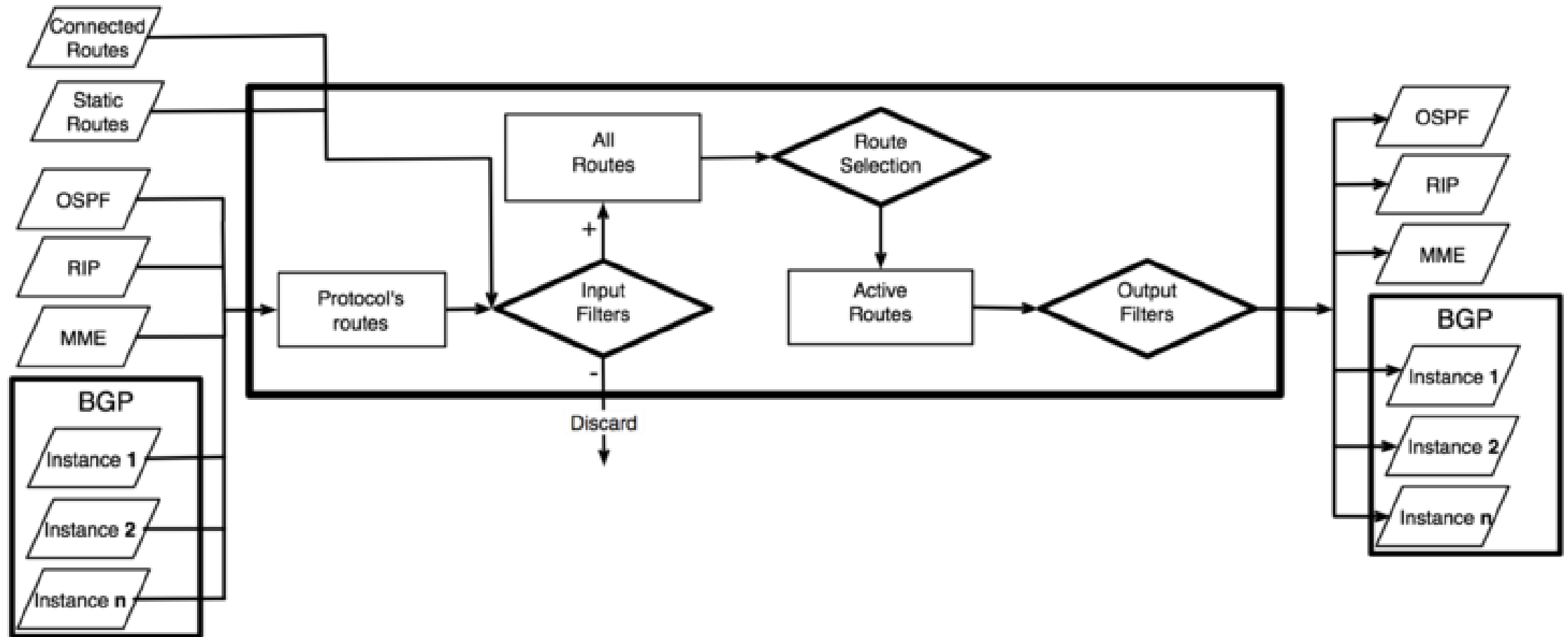
MikroTik RouterOS & RouterBOARD

- MikroTik adalah sebuah perusahaan perangkat keras jaringan yang berada di Riga, Latvia, Eropa yang berdiri pada tahun 1995
- MikroTik menggunakan Kernel Linux 2.2 pada saat pertama kali membuat RouterOS
- RouterOS adalah sebuah Operating System yang dibuat oleh MikroTik dan biasanya digunakan kepada PC
- RouterBOARD adalah sebuah perangkat jaringan (router) yang menggunakan RouterOS
- Fitur MikroTik: Routing, Firewall, NAT, Hotspot, QoS, Tunnel, DHCP, MNDP, DNS, Web Proxy, Wireless, Bridging, Monitoring

Routing

- Routing adalah suatu proses pemilihan jalur (path) yang akan dilalui sebuah paket dan dilakukan oleh sebuah perangkat keras jaringan yang bernama router
- Routing Process memiliki 2 tahapan yang cukup rumit yaitu pada Routing Information Base (RIB) dan Forwarding Information Base (FIB)
- RIB atau biasa disebut Routing Table menyimpan dan mengenerate path secara statis ataupun dinamis dan melakukan filtering juga
- FIB memiliki tugas untuk menyimpan cache dari RIB

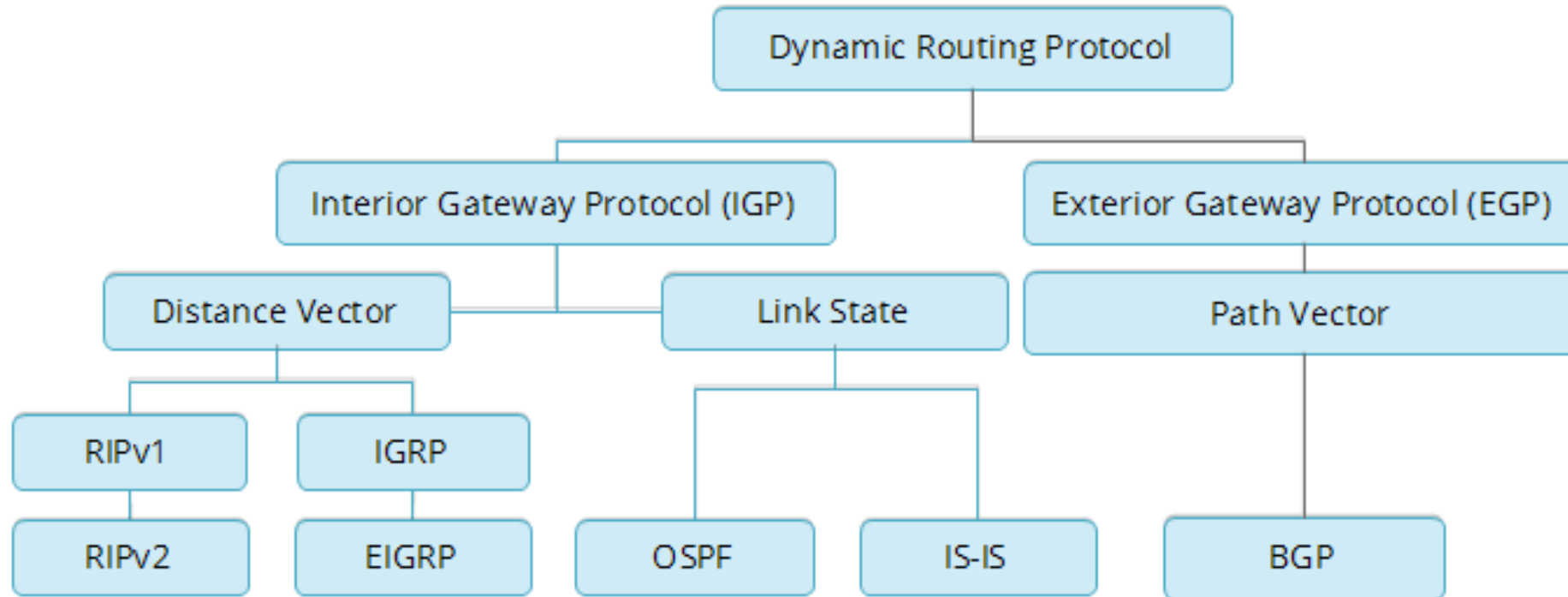
Routing Information Base (RIB)



Images was taken from mikrotik.com

Routing Protocols

- Static Routing



Images was taken from google images

Open Shortest Path First (OSPF)

- Open Shortest Path First atau disingkat menjadi OSPF adalah salah satu dari Dynamic Routing Protocol yang ada dan banyak digunakan saat ini karena memiliki fitur yang bisa dikembangkan dan algoritma berdasarkan kecepatan, bukan lagi hop count seperti RIP
- Fitur OSPF: Standard Area, Stub Area, Not So Stubby Area (NSSA), Area Authentication, Redistribute, Virtual-Link, DR & DBR (Designated Router & Designated Backup Router)

OSPF Request for Comments (RFC)

- OSPFv2 – RFC 2328 (IPv4)
- OSPFv3 – RFC 5304 (IPv6)

Kelebihan (OSPF)

- OSPF adalah IGP (Interior Gateway Protocol)
- Berkategori Link-State
- Tidak ada batasan hop count
- Komunikasi melalui multicast (224.0.0.5)
- Update jika ada perubahan topologi
- Bisa membagi router menjadi beberapa area
- Bisa otomatis membagi default route

Kekurangan OSPF

- Membutuhkan resource lebih (CPU & Memory)
- Lebih kompleks dan sulit dari pada RIP

Istilah dalam OSPF

- **Neighbor** - Router lain yang telah Adjacency
- **Adjacency** - Koneksi antar router secara logikal
- **Link** - Hubungan antar Router
- **Interface** - Physical Interface
- **LSA** - Link State Advertisement, data yang dikirimkan oleh OSPF
- **DR** - Designated Router, router yang dipilih untuk mengumpulkan LSA jika network-type=NBMA
- **BDR** -Backup Designated Router, cadangan dari DR dan menggunakan IP multicast 224.0.0.6 (antar DR dan BDR)
- **Area** - areas digunakan untuk melakukan segmentasi, agar flooding paket LSA tidak terlalu besar
- **ABR** - Area Border Router, router yang terkoneksi dengan area lain
- **ASBR** - Autonomous System Boundary Router, router yang terkoneksi ke network luar

Istilah dalam OSPF

- **NBMA** - Non-broadcast multi-access, mengumpulkan LSA kedalam DR
- **Broadcast** - Network yang membolehkan broadcast, contohnya ethernet
- **Point-to-point** - Network type yang tidak memerlukan DR dan BDR karean hanya PTP
- **Router-ID** - IP address yang digunakan untuk mengidentifikasi Router OSPF
- **Link State** - Status dari kondisi antar router
- **Cost** - Acuan untuk memilih path
- **Autonomous System Number** - Autonomous System Number (ASN) adalah suatu nomor identitas dari sekumpulan perangkat jaringan (contoh: ISP, Kampus, Sekolah, Perusahaan)

Cara Kerja OSPF

- Akan mencari jalur terbaik berdasarkan cost dengan menggunakan rumus $100\text{mbps}/\text{interface bandwidth (mb)}$, contoh suatu koneksi ke network B memiliki bandwidth interface sebesar 100mbps, maka perhitungan cost adalah $100/100 = 1$
- Makin kecil cost, maka akan semakin dipilih sebagai jalur
- Flooding LSA (Link-State Advertisements) LSA merupakan informasi dari network, area, authentication dan router-id sebuah router dan melakukan mapping topology juga setelah mendapatkan semua informasi dan akan melakukan kalkulasi ulang jika ada perubahan topology, maka dari itu dibuat segmentasi (area) agar flooding LSA tidak terlalu membebani jaringan

Cara Kerja OSPF

- Kirim Hello Packet (Kenalan Dulu)
- Kirim Database Description (DBD dari routing table yang dimiliki)
- Kirim LSR (Link-State Request, masing – masing router akan meminta network dari kawan – kawannya)
- Kirim LSU (Link-State Update, mengirim network yang dimiliki)
- Kirim LSAck (Link-State Acknowledge, mengkonfirmasi bahwa packet telah tiba dengan selamat)

Jenis – Jenis Router OSPF

- ASBR (Autonomous System Boundary Router): sebuah router yang terhubung dengan routing protocol lain atau ke gateway internet
- ABR (Area Border Router): sebuah router yang berada di tengah – tengah dari 2 area
- IR (Internal Router): router yang berada pada sebuah area, maksimal Internal Router pada sebuah area adalah 79 IR + 1 ABR, namun dalam dunia nyata, ada baiknya membuat area baru jika sudah 50 router

OSPF Network-Type

- Broadcast
- Non-Broadcast Multiple Access (NBMA) : Mengumpulkan LSA pada Designated Router (DR) dan Backup Designated Router (BDR)
- Point-to-Point
- Point-to-Multipoint (PTMP)

*Network yang ditangani, bukan cara komunikasi

OSPF LSA Type

- type 1 - (Router LSA) Sent by routers within the Area, including the list of directly attached links. Does not cross the ABR or ASBR.
- type 2 - (Network LSA) Generated for every “transit network” within an area. A transit network has at least two directly attached OSPF routers. Ethernet is an example of a Transit Network. A Type 2 LSA lists each of the attached routers that make up the transit network and is generated by the DR.
- type 3 - (Summary LSA) The ABR sends Type 3 Summary LSAs. A Type 3 LSA advertises any networks owned by an area to the rest of the areas in the OSPF AS. By default, OSPF advertises Type 3 LSAs for every subnet defined in the originating area, which can cause flooding problems, so it’s a good idea to use a manual summarization at the ABR.
- type 4 - (ASBR-Summary LSA) It announces the ASBR address, it shows “where” the ASBR is located, announcing it’s address instead of it’s routing table.
- type 5 - (External LSA) Announces the Routes learned through the ASBR. External LSAs are flooded to all areas except Stub areas. These LSAs divides in two types: external type 1 and external type2.
- type 6 - (Group Membership LSA) This was defined for Multicast extensions to OSPF and is not used by ROuterOS.
- type 7 - type 7 LSAs are used to tell the ABRs about these external routes imorted in NSSA area. Area Border Router then translates these LSAs to type 5 external LSAs and floods as normal to the rest of the OSPF network
- type 8 - (Link-local only LSA for OSPFv3)

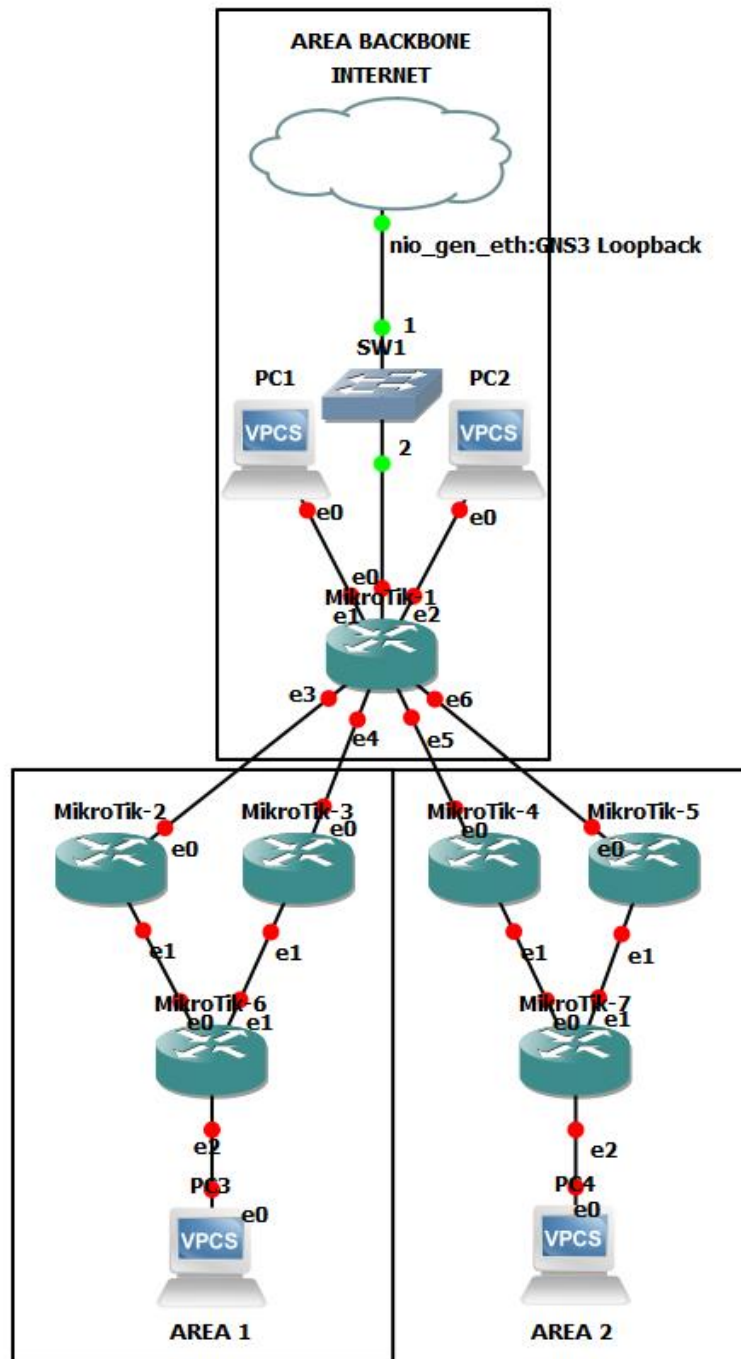
Hello Data

- Netmask
- Hello Interval
- Options & Priority
- Dead Interval
- DR
- BDR
- 1st Neighbor ID

OSPF Header

- Version & Packet Type
- Length
- Router ID
- Area ID
- Packet Checksum
- Authentication Type
- Authentication Data

Topologi OSPF



Konfigurasi Cloud

- Tambahkan GNS3 Loopback Adapter, dan hubungkan kepada switch

Konfigurasi Router MikroTik-1

- Lakukan Bridging antara ether2 dan ether3 untuk NOC, dan buatlah Loopback adapter dan berikan identitas

```
/system identity set name=R1
```

```
/interface bridge
```

```
add name=NOC
```

```
add name=loopback
```

```
/interface bridge port
```

```
add bridge=NOC interface=ether2
```

```
add bridge=NOC interface=ether3
```

Konfigurasi Router MikroTik-1

- Berikan IP Address Pada setiap Interface

/ip address

add address=1.1.1.1 interface=loopback

add address=10.10.10.1/29 interface=NOC

add address=192.168.137.100/24 interface=ether1

add address=172.16.10.1/30 interface=ether4

add address=172.16.20.1/30 interface=ether5

add address=172.16.30.1/30 interface=ether6

add address=172.16.40.1/30 interface=ether7

Konfigurasi Router MikroTik-1

- Berikan DHCP Server dan DNS pada interface NOC agar PC NOC mendapatkan IP secara otomatis

```
/ip dns set allow-remote-requests=yes servers=192.168.137.1
```

```
/ip pool add name=dhcp_pool1 ranges=10.10.10.2-10.10.10.6
```

```
/ip dhcp-server
```

```
add address-pool=dhcp_pool1 disabled=no interface=NOC  
name=dhcp1
```

```
/ip dhcp-server network
```

```
add address=10.10.10.0/29 dns-server=192.168.137.1  
gateway=10.10.10.1
```


Konfigurasi Router MikroTik-1

- Lakukan NAT dan static route kepada internet

```
/ip firewall nat
```

```
add action=masquerade chain=srcnat out-interface=ether1
```

```
/ip route gateway=192.168.137.1
```

Konfigurasi Router MikroTik-1

- Berikan router-id dan redistribute default route

/routing ospf area

add area-id=0.0.0.1 name=area1

add area-id=0.0.0.2 name=area2

/routing ospf instance

set distribute-default=always-as-type-1 router-id=1.1.1.1

Konfigurasi Router MikroTik-1

- Tambahkan Network yang ingin di advertise dan password router
/routing ospf network

add area=backbone network=1.1.1.1/32

add area=backbone network=10.10.10.0/29

add area=backbone network=192.168.137.0/24

add area=area1 network=172.16.10.0/30

add area=area1 network=172.16.20.0/30

add area=area2 network=172.16.30.0/30

add area=area2 network=172.16.40.0/30

/user set admin password=laporan

Konfigurasi Router MikroTik-2

- Konfigurasi Loopback adapter, area, router-id, DNS dan IP Address dan identitas

```
/system identity set name=R2
```

```
/interface bridge add name=loopback
```

```
/routing ospf area add area-id=0.0.0.1 name=area1
```

```
/routing ospf instance set router-id=2.2.2.2
```

```
/ip dns set allow-remote-requests=yes servers=192.168.137.1
```

```
/ip address add address=2.2.2.2 interface=loopback
```

```
/ip address add address=172.16.10.2/30 interface=ether1
```

```
/ip address add address=192.168.10.1/30 interface=ether2
```

Konfigurasi Router MikroTik-2

- Tambahkan network yang ingin di advertise dan password router

/routing ospf network

add area=area1 network=2.2.2.2/32

add area=area1 network=172.16.10.0/30

add area=area1 network=192.168.10.0/30

/user set admin password=laporan

Konfigurasi Router MikroTik-3

- Konfigurasi Loopback adapter, area, router-id, DNS dan IP Address dan identitas

```
/system identity set name=R3
```

```
/interface bridge add name=loopback
```

```
/routing ospf area add area-id=0.0.0.1 name=area1
```

```
/routing ospf instance set router-id=3.3.3.3
```

```
/ip dns set allow-remote-requests=yes servers=192.168.137.1
```

```
/ip address add address=3.3.3.3 interface=loopback
```

```
/ip address add address=172.16.20.2/30 interface=ether1
```

```
/ip address add address=192.168.20.1/30 interface=ether2
```

Konfigurasi Router MikroTik-3

- Tambahkan network yang ingin di advertise dan password router

/routing ospf network

add area=area1 network=3.3.3.3/32

add area=area1 network=172.16.20.0/30

add area=area1 network=192.168.20.0/30

/user set admin password=laporan

Konfigurasi Router MikroTik-4

- Konfigurasi Loopback adapter, area, router-id, DNS dan IP Address dan identitas

```
/system identity set name=R4
```

```
/interface bridge add name=loopback
```

```
/routing ospf area add area-id=0.0.0.2 name=area2
```

```
/routing ospf instance set router-id=4.4.4.4
```

```
/ip dns set allow-remote-requests=yes servers=192.168.137.1
```

```
/ip address add address=4.4.4.4 interface=loopback
```

```
/ip address add address=172.16.30.2/30 interface=ether1
```

```
/ip address add address=192.168.30.1/30 interface=ether2
```


Konfigurasi Router MikroTik-4

- Tambahkan network yang ingin di advertise dan password router

/routing ospf network

add area=area2 network=3.3.3.3/32

add area=area2 network=172.16.20.0/30

add area=area2 network=192.168.20.0/30

/user set admin password=laporan

Konfigurasi Router MikroTik-5

- Konfigurasi Loopback adapter, area, router-id, DNS dan IP Address dan identitas

```
/system identity set name=R5
```

```
/interface bridge add name=loopback
```

```
/routing ospf area add area-id=0.0.0.2 name=area2
```

```
/routing ospf instance set router-id=5.5.5.5
```

```
/ip dns set allow-remote-requests=yes servers=192.168.137.1
```

```
/ip address add address=5.5.5.5 interface=loopback
```

```
/ip address add address=172.16.40.2/30 interface=ether1
```

```
/ip address add address=192.168.40.1/30 interface=ether2
```

Konfigurasi Router MikroTik-5

- Tambahkan network yang ingin di advertise dan password router

/routing ospf network

add area=area2 network=5.5.5.5/32

add area=area2 network=172.16.40.0/30

add area=area2 network=192.168.40.0/30

/user set admin password=laporan

Konfigurasi Router MikroTik-6

- Konfigurasi area, router-id dan IP Address dan juga identitas

```
/system identity set name=R6
```

```
/routing ospf area add area-id=0.0.0.1 name=area1
```

```
/routing ospf instance set router-id=6.6.6.6
```

```
/ip address
```

```
add address=192.168.10.2/30 interface=ether1
```

```
add address=192.168.20.2/30 interface=ether2
```

```
add address=100.100.100.1/24 interface=ether3
```

```
add address=6.6.6.6 interface=loopback
```

Konfigurasi Router MikroTik-6

- Konfigurasi Loopback Adapter dan DHCP Server dan DNS

```
/interface bridge add name=loopback
```

```
/ip dns set allow-remote-requests=yes servers=192.168.137.1
```

```
/ip pool
```

```
add name=dhcp_pool1 ranges=100.100.100.2-100.100.100.254
```

```
/ip dhcp-server
```

```
add address-pool=dhcp_pool1 disabled=no interface=ether3 name=dhcp1
```

```
/ip dhcp-server network
```

```
add address=100.100.100.0/24 dns-server=192.168.137.1
```

```
gateway=100.100.100.1
```

Konfigurasi Router MikroTik-6

Tambahkan network yang ingin di Advertise

/routing ospf network

add area=area1 network=192.168.10.0/30

add area=area1 network=192.168.20.0/30

add area=area1 network=6.6.6.6/32

add area=area1 network=100.100.100.0/24

Konfigurasi Router MikroTik-7

- Konfigurasi area, router-id dan IP Address dan juga identitas

```
/system identity set name=R7
```

```
/routing ospf area add area-id=0.0.0.2 name=area2
```

```
/routing ospf instance set router-id=7.7.7.7
```

```
/ip address
```

```
add address=192.168.30.2/30 interface=ether1
```

```
add address=192.168.40.2/30 interface=ether2
```

```
add address=200.200.200.1/24 interface=ether3
```

```
add address=7.7.7.7 interface=loopback
```

Konfigurasi Router MikroTik-7

- Konfigurasi Loopback Adapter dan DHCP Server dan DNS

```
/interface bridge add name=loopback
```

```
/ip dns set allow-remote-requests=yes servers=192.168.137.1
```

```
/ip pool
```

```
add name=dhcp_pool1 ranges=200.200.200.2-200.200.200.254
```

```
/ip dhcp-server
```

```
add address-pool=dhcp_pool1 disabled=no interface=ether3 name=dhcp1
```

```
/ip dhcp-server network
```

```
add address=200.200.200.0/24 dns-server=192.168.137.1
```

```
gateway=200.200.200.1
```


Konfigurasi Router MikroTik-7

- Tambahkan network yang ingin di Advertise

/routing ospf network

```
add area=area1 network=192.168.30.0/30
```

```
add area=area1 network=192.168.40.0/30
```

```
add area=area1 network=7.7.7.7/32
```

```
add area=area1 network=200.200.200.0/24
```

Konfigurasi PC

- Konfigurasikan semua IP Address PC menjadi DHCP

PC1> dhcp

PC2> dhcp

PC3> dhcp

PC4> dhcp

Finishing

- Check semua Routing Table setiap router menggunakan perintah
 /ip route print
- Lalu perhatikan, apakah semua network sudah masuk dan memiliki flag Active Dynamic OSPF (ADo) dan jika network sendiri flagnya Active Dynamic Connected (ADC)

MikroTik-1 & MikroTik-2 (/ip route print)

```
MikroTik-1
:
: tonum typeof
environment do find global local pick return toarray toip
:tostr while
[admin@M1] >
caps-man disk file ipv6 port radius special-login user impo
rt quit undo
certificate driver interface log ppp routing system beep pass
word redo
console dude ip mpls queue snmp tool export ping
setup
[admin@M1] > ip route print
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b
- bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 A S 0.0.0.0/0 192.168.137.1 1
1 ADC 1.1.1.1/32 1.1.1.1 loopback 0
2 ADo 2.2.2.2/32 172.16.10.2 110
3 ADo 3.3.3.3/32 172.16.20.2 110
4 ADo 4.4.4.4/32 172.16.30.2 110
5 ADo 5.5.5.5/32 172.16.40.2 110
6 ADo 6.6.6.6/32 172.16.20.2 110
7 ADo 7.7.7.7/32 172.16.30.2 110
8 ADC 10.10.10.0/29 10.10.10.1 NOC 0
9 ADo 100.100.100.0/24 172.16.20.2 110
10 ADC 172.16.10.0/30 172.16.10.1 ether4 0
11 ADC 172.16.20.0/30 172.16.20.1 ether5 0
12 ADC 172.16.30.0/30 172.16.30.1 ether6 0
13 ADC 172.16.40.0/30 172.16.40.1 ether7 0
14 ADo 192.168.10.0/30 172.16.10.2 110
15 ADo 192.168.20.0/30 172.16.20.2 110
16 ADo 192.168.30.0/30 172.16.30.2 110
17 ADo 192.168.40.0/30 172.16.40.2 110
18 ADC 192.168.137.0/24 192.168.137.100 ether1 0
19 ADo 200.200.200.0/24 172.16.30.2 110
[admin@M1] >

MikroTik-2
[admin@M2] >
caps-man disk file ipv6 port radius special-login user impo
rt quit undo
certificate driver interface log ppp routing system beep pass
word redo
console dude ip mpls queue snmp tool export ping
setup
[admin@M2] >
/ terminal error for if nothing put set tobool
toip6 totime
: delay execute foreach len parse resolve time toid
: tonum typeof
environment do find global local pick return toarray toip
:tostr while
[admin@M2] > ip route print
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b
- bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 0.0.0.0/0 172.16.10.1 110
1 ADo 1.1.1.1/32 172.16.10.1 110
2 ADC 2.2.2.2/32 2.2.2.2 loopback 0
3 ADo 3.3.3.3/32 192.168.10.2 110
4 ADo 4.4.4.4/32 172.16.10.1 110
5 ADo 5.5.5.5/32 172.16.10.1 110
6 ADo 6.6.6.6/32 192.168.10.2 110
7 ADo 7.7.7.7/32 172.16.10.1 110
8 ADo 10.10.10.0/29 172.16.10.1 110
9 ADo 100.100.100.0/24 192.168.10.2 110
10 ADC 172.16.10.0/30 172.16.10.2 ether1 0
11 ADo 172.16.20.0/30 172.16.10.1 110
12 ADo 172.16.30.0/30 172.16.10.1 110
13 ADo 172.16.40.0/30 172.16.10.1 110
14 ADC 192.168.10.0/30 192.168.10.1 ether2 0
15 ADo 192.168.20.0/30 192.168.10.2 110
16 ADo 192.168.30.0/30 172.16.10.1 110
17 ADo 192.168.40.0/30 172.16.10.1 110
18 ADo 192.168.137.0/24 172.16.10.1 110
19 ADo 200.200.200.0/24 172.16.10.1 110
[admin@M2] >
```

MikroTik-3 & MikroTik-4 (/ip route print)

```
[admin@R3] >
/
toip6 totime
: delay execute foreach len parse resolve time toid
tonum typeof
environment do find global local pick return toarray toip
tostr while
[admin@R3] >
caps-man disk file ipv6 port radius special-login user impo
rt quit undo
certificate driver interface log ppp routing system beep pass
word redo
console dude ip mpls queue snmp tool export ping
setup
[admin@R3] > ip route print
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b
- bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 0.0.0.0/0 172.16.20.1 110
1 ADo 1.1.1.1/32 172.16.20.1 110
2 ADo 2.2.2.2/32 172.16.20.1 110
3 ADC 3.3.3.3/32 3.3.3.3 loopback 0
4 ADo 4.4.4.4/32 172.16.20.1 110
5 ADo 5.5.5.5/32 172.16.20.1 110
6 ADo 6.6.6.6/32 192.168.20.2 110
7 ADo 7.7.7.7/32 172.16.20.1 110
8 ADo 10.10.10.0/29 172.16.20.1 110
9 ADo 100.100.100.0/24 192.168.20.2 110
10 ADo 172.16.10.0/30 172.16.20.1 110
11 ADC 172.16.20.0/30 172.16.20.2 ether1 0
12 ADo 172.16.30.0/30 172.16.20.1 110
13 ADo 172.16.40.0/30 172.16.20.1 110
14 ADo 192.168.10.0/30 192.168.20.2 110
15 ADC 192.168.20.0/30 192.168.20.1 ether2 0
16 ADo 192.168.30.0/30 172.16.20.1 110
17 ADo 192.168.40.0/30 172.16.20.1 110
18 ADo 192.168.137.0/24 172.16.20.1 110
19 ADo 200.200.200.0/24 172.16.20.1 110
[admin@R3] > []
```

```
[admin@R4] >
caps-man driver ip port routing tool import redo
certificate dude ipv6 ppp snmp user password setup
console file log queue special-login beep ping undo
disk interface mpls radius system export quit
[admin@R4] >
/
delay find if parse return tobool tonum while
: do for len pick set toid tostr
environment error foreach local put time toip totime
terminal execute global nothing resolve toarray toip6 typeof
[admin@R4] >
caps-man driver ip port routing tool import redo
certificate dude ipv6 ppp snmp user password setup
console file log queue special-login beep ping undo
disk interface mpls radius system export quit
[admin@R4] > ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 0.0.0.0/0 172.16.30.1 110
1 ADo 1.1.1.1/32 172.16.30.1 110
2 ADo 2.2.2.2/32 172.16.30.1 110
3 ADo 3.3.3.3/32 172.16.30.1 110
4 ADC 4.4.4.4/32 4.4.4.4 loopback 0
5 ADo 5.5.5.5/32 172.16.30.1 110
6 ADo 6.6.6.6/32 192.168.30.2 110
7 ADo 7.7.7.7/32 192.168.30.2 110
8 ADo 10.10.10.0/29 172.16.30.1 110
9 ADo 100.100.100.0/24 172.16.30.1 110
10 ADo 172.16.10.0/30 172.16.30.1 110
11 ADo 172.16.20.0/30 172.16.30.1 110
12 ADC 172.16.30.0/30 172.16.30.2 ether1 0
13 ADo 172.16.40.0/30 172.16.30.1 110
14 ADo 192.168.10.0/30 172.16.30.1 110
15 ADo 192.168.20.0/30 172.16.30.1 110
16 ADC 192.168.30.0/30 192.168.30.1 ether2 0
17 ADo 192.168.40.0/30 192.168.30.2 110
18 ADo 192.168.137.0/24 172.16.30.1 110
19 ADo 200.200.200.0/24 192.168.30.2 110
[admin@R4] > []
```

MikroTik-5 & MikroTik-6 (/ip route print)

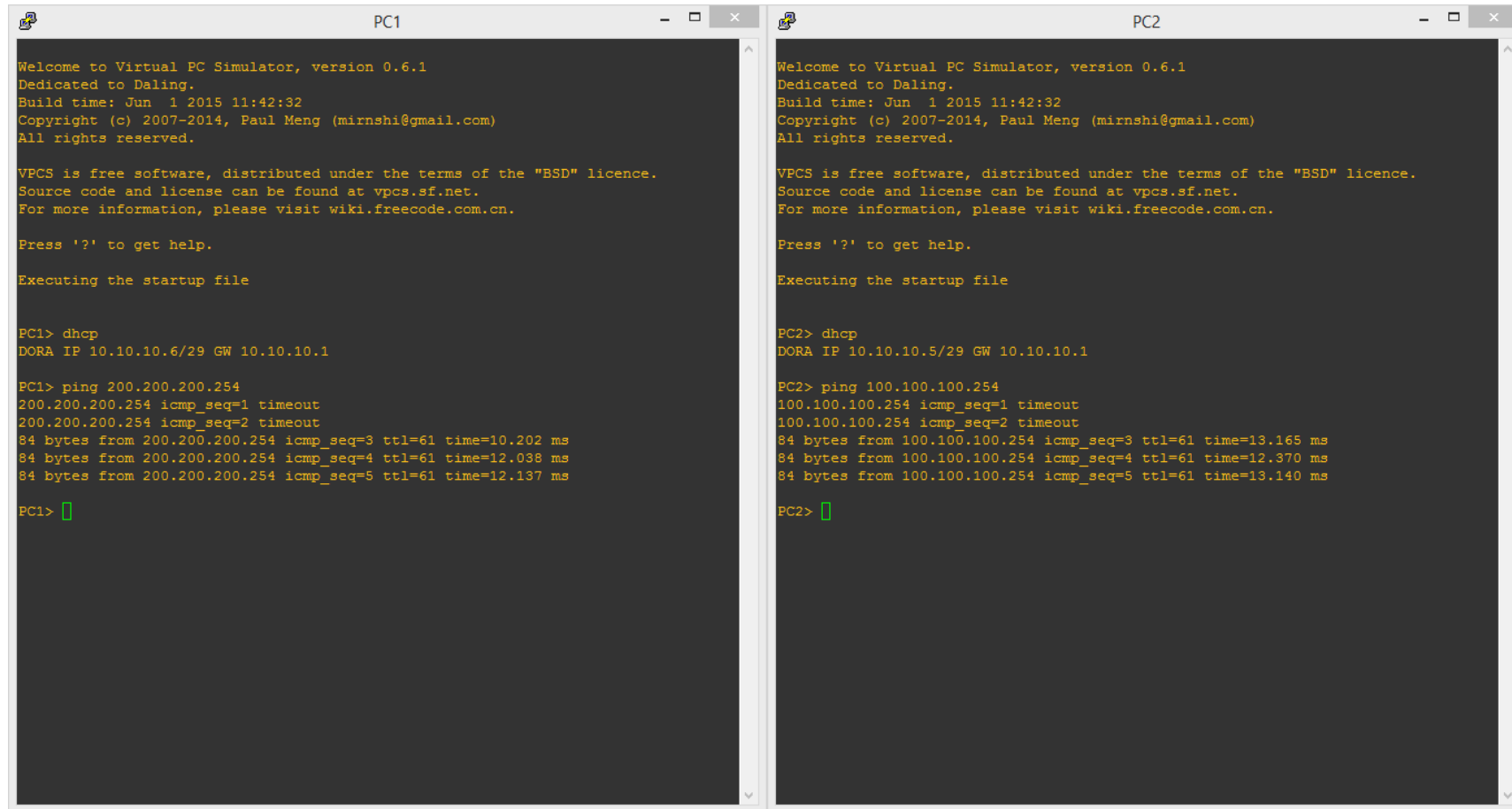
```
[admin@M5] >
caps-man disk file ipv6 port radius special-login user impo
rt quit undo
certificate driver interface log ppp routing system beep pass
word redo
console dude ip mpls queue snmp tool export ping
setup
[admin@M5] >
/ terminal error for if nothing put set tobool
toip6 totime
: delay execute foreach len parse resolve time toid
tonum typeof
environment do find global local pick return toarray toip
tostr while
[admin@M5] > ip route print
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b
- bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 0.0.0.0/0 192.168.20.1 110
1 ADo 1.1.1.1/32 192.168.20.1 110
2 ADo 2.2.2.2/32 192.168.20.1 110
3 ADo 3.3.3.3/32 192.168.20.1 110
4 ADo 4.4.4.4/32 192.168.20.1 110
5 ADC 5.5.5.5/32 5.5.5.5 loopback 0
6 ADo 6.6.6.6/32 172.16.40.1 110
7 ADo 7.7.7.7/32 192.168.40.2 110
8 ADo 10.10.10.0/29 172.16.40.1 110
9 ADo 100.100.100.0/24 172.16.40.1 110
10 ADo 172.16.10.0/30 172.16.40.1 110
11 ADo 172.16.20.0/30 172.16.40.1 110
12 ADo 172.16.30.0/30 172.16.40.1 110
13 ADC 172.16.40.0/30 172.16.40.2 ether1 0
14 ADo 192.168.10.0/30 172.16.40.1 110
15 ADo 192.168.20.0/30 172.16.40.1 110
16 ADo 192.168.30.0/30 192.168.40.2 110
17 ADC 192.168.40.0/30 192.168.40.1 ether2 0
18 ADo 192.168.137.0/24 172.16.40.1 110
19 ADo 200.200.200.0/24 192.168.40.2 110
[admin@M5] > []
```

```
[admin@M6] >
tonum typeof
environment do find global local pick return toarray toip
tostr while
[admin@M6] > ip route print
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b
- bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 0.0.0.0/0 192.168.20.1 110
1 ADo 1.1.1.1/32 192.168.20.1 110
2 ADo 2.2.2.2/32 192.168.20.1 110
3 ADo 3.3.3.3/32 192.168.20.1 110
4 ADo 4.4.4.4/32 192.168.20.1 110
5 ADo 5.5.5.5/32 192.168.20.1 110
6 ADC 6.6.6.6/32 6.6.6.6 loopback 0
7 ADo 7.7.7.7/32 192.168.20.1 110
8 ADo 10.10.10.0/29 192.168.20.1 110
9 ADC 100.100.100.0/24 100.100.100.1 ether3 0
10 ADo 172.16.10.0/30 192.168.10.1 110
11 ADo 172.16.20.0/30 192.168.20.1 110
12 ADo 172.16.30.0/30 192.168.20.1 110
13 ADo 172.16.40.0/30 192.168.10.1 110
14 ADC 192.168.10.0/30 192.168.10.2 ether1 0
15 ADC 192.168.20.0/30 192.168.20.2 ether2 0
16 ADo 192.168.30.0/30 192.168.20.1 110
17 ADo 192.168.40.0/30 192.168.20.1 110
18 ADo 192.168.137.0/24 192.168.10.1 110
19 ADo 200.200.200.0/24 192.168.20.1 110
[admin@M6] > []
```

MikroTik-7 (/ip route print)

```
MikroTik-7
caps-man  console  driver  file  ip  log  port  queue  routing  special-login  tool  beep  import  ping  redo  undo
certificate disk  dude  interface ipv6 mpls ppp radius snmp  system  user  export  password  quit  setup
[admin@M7] >
/  environment  delay  error  find  foreach  if  local  parse  put  return  time  tobool  toip  tonum  totime  while
:  terminal  do  execute  for  global  len  nothing  pick  resolve  set  toarray  toid  toip6  tostr  typeof
[admin@M7] >
caps-man  console  driver  file  ip  log  port  queue  routing  special-login  tool  beep  import  ping  redo  undo
certificate disk  dude  interface ipv6 mpls ppp radius snmp  system  user  export  password  quit  setup
[admin@M7] > ip route print
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme, B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 ADo  0.0.0.0/0           192.168.40.1  110
1 ADo  1.1.1.1/32          192.168.40.1  110
2 ADo  2.2.2.2/32          192.168.40.1  110
3 ADo  3.3.3.3/32          192.168.40.1  110
4 ADo  4.4.4.4/32          192.168.40.1  110
5 ADo  5.5.5.5/32          192.168.40.1  110
6 ADo  6.6.6.6/32          192.168.40.1  110
7 ADC  7.7.7.7/32          7.7.7.7      loopback      0
8 ADo  10.10.10.0/29        192.168.40.1  110
9 ADo  100.100.100.0/24     192.168.40.1  110
10 ADo  172.16.10.0/30       192.168.40.1  110
11 ADo  172.16.20.0/30       192.168.40.1  110
12 ADo  172.16.30.0/30       192.168.40.1  110
13 ADo  172.16.40.0/30       192.168.40.1  110
14 ADo  192.168.10.0/30      192.168.40.1  110
15 ADo  192.168.20.0/30      192.168.40.1  110
16 ADC  192.168.30.0/30      192.168.30.2  ether1         0
17 ADC  192.168.40.0/30      192.168.40.2  ether2         0
18 ADo  192.168.137.0/24     192.168.40.1  110
19 ADC  200.200.200.0/24     200.200.200.1 ether3         0
[admin@M7] > []
```

PC1 & PC2 (DHCP & PING)



The image shows two side-by-side terminal windows from a 'Virtual PC Simulator, version 0.6.1'. Both windows display the same introductory text: 'Welcome to Virtual PC Simulator, version 0.6.1', 'Dedicated to Daling.', 'Build time: Jun 1 2015 11:42:32', 'Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)', 'All rights reserved.', 'VPCS is free software, distributed under the terms of the "BSD" licence.', 'Source code and license can be found at vpcs.sf.net.', 'For more information, please visit wiki.freecode.com.cn.', 'Press '?' to get help.', and 'Executing the startup file'. The PC1 window shows the command 'PC1> dhcp' followed by 'DORA IP 10.10.10.6/29 GW 10.10.10.1', and then a series of ping commands to 200.200.200.254, with results showing timeouts for the first two and successful responses for the next three. The PC2 window shows the command 'PC2> dhcp' followed by 'DORA IP 10.10.10.5/29 GW 10.10.10.1', and then a series of ping commands to 100.100.100.254, with results showing timeouts for the first two and successful responses for the next three.

```
PC1
Welcome to Virtual PC Simulator, version 0.6.1
Dedicated to Daling.
Build time: Jun 1 2015 11:42:32
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> dhcp
DORA IP 10.10.10.6/29 GW 10.10.10.1

PC1> ping 200.200.200.254
200.200.200.254 icmp_seq=1 timeout
200.200.200.254 icmp_seq=2 timeout
84 bytes from 200.200.200.254 icmp_seq=3 ttl=61 time=10.202 ms
84 bytes from 200.200.200.254 icmp_seq=4 ttl=61 time=12.038 ms
84 bytes from 200.200.200.254 icmp_seq=5 ttl=61 time=12.137 ms

PC1>

PC2
Welcome to Virtual PC Simulator, version 0.6.1
Dedicated to Daling.
Build time: Jun 1 2015 11:42:32
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

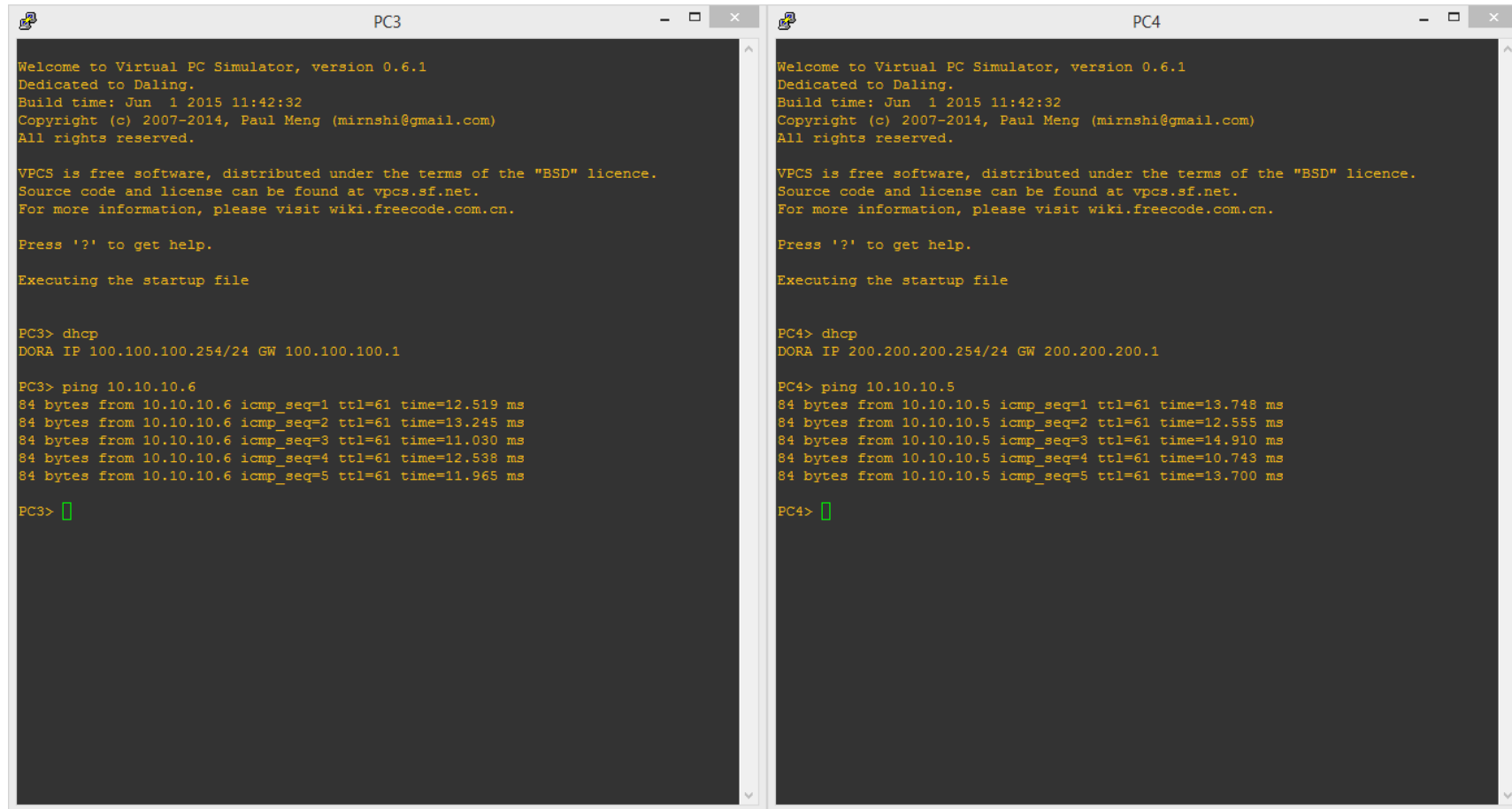
Executing the startup file

PC2> dhcp
DORA IP 10.10.10.5/29 GW 10.10.10.1

PC2> ping 100.100.100.254
100.100.100.254 icmp_seq=1 timeout
100.100.100.254 icmp_seq=2 timeout
84 bytes from 100.100.100.254 icmp_seq=3 ttl=61 time=13.165 ms
84 bytes from 100.100.100.254 icmp_seq=4 ttl=61 time=12.370 ms
84 bytes from 100.100.100.254 icmp_seq=5 ttl=61 time=13.140 ms

PC2>
```


PC3 & PC4 (DHCP & PING)



The image displays two side-by-side terminal windows from a Virtual PC Simulator. The left window, titled 'PC3', shows the execution of a DHCP command and a ping test to 10.10.10.6. The right window, titled 'PC4', shows the execution of a DHCP command and a ping test to 10.10.10.5. Both windows display the same introductory text about the simulator version and license.

```
PC3
Welcome to Virtual PC Simulator, version 0.6.1
Dedicated to Daling.
Build time: Jun  1 2015 11:42:32
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> dhcp
DORA IP 100.100.100.254/24 GW 100.100.100.1

PC3> ping 10.10.10.6
84 bytes from 10.10.10.6 icmp_seq=1 ttl=61 time=12.519 ms
84 bytes from 10.10.10.6 icmp_seq=2 ttl=61 time=13.245 ms
84 bytes from 10.10.10.6 icmp_seq=3 ttl=61 time=11.030 ms
84 bytes from 10.10.10.6 icmp_seq=4 ttl=61 time=12.538 ms
84 bytes from 10.10.10.6 icmp_seq=5 ttl=61 time=11.965 ms

PC3>

PC4
Welcome to Virtual PC Simulator, version 0.6.1
Dedicated to Daling.
Build time: Jun  1 2015 11:42:32
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> dhcp
DORA IP 200.200.200.254/24 GW 200.200.200.1

PC4> ping 10.10.10.5
84 bytes from 10.10.10.5 icmp_seq=1 ttl=61 time=13.748 ms
84 bytes from 10.10.10.5 icmp_seq=2 ttl=61 time=12.555 ms
84 bytes from 10.10.10.5 icmp_seq=3 ttl=61 time=14.910 ms
84 bytes from 10.10.10.5 icmp_seq=4 ttl=61 time=10.743 ms
84 bytes from 10.10.10.5 icmp_seq=5 ttl=61 time=13.700 ms

PC4>
```

Contact Me

- Facebook: <https://www.facebook.com/mict404>
(Michael Takeuchi)
- LinkedIn: <https://www.linkedin.com/in/michael-takeuchi>
(Michael Takeuchi)
- Twitter: [@mict404](https://twitter.com/mict404)
- Blog: <http://michael-takeuchi.blogspot.com>

Thank You !